Bank Loan Performance Dashboard

Project problem statement, business glossary, KPI definitions and analytical scope

1. Problem statement

A lending team needs a consistent way to monitor loan origination, funding, repayment activity, borrower profiles and loan outcomes. Reviewing individual records makes it difficult to see portfolio-level trends or distinguish loans in good standing from loans that have been charged off. This project organizes loan-level data into Excel dashboards that summarize lending activity and support portfolio monitoring and exploratory risk analysis.

2. Project objectives

- Measure the volume and value of loan applications and funded loans.
- Monitor payments received, average interest rates and borrower debt-to-income ratios.
- Compare loan outcomes, including fully paid, current and charged-off loans.
- Explore variation by issue month, geography, purpose, term, employment length and home ownership.
- Provide repeatable summary and overview dashboards for filtering and reporting.

3. Workbook inventory and scope

The workbook contains four worksheets: Summary Dashboard, Overview Dashboard, Sheet Design and financial_loan. The financial_loan table contains 38,576 data rows and 25 columns, including a derived Good Loan vs Bad Loan classification. The Sheet Design worksheet contains supporting KPI calculations. The data source, extraction date, currency and institutional ownership are not documented in the workbook and should be confirmed before public publication.

4. Headline measures from workbook

- Loan applications: 38,576
- Funded amount: 435,757,075 (currency unspecified)
- Total payments received: 473,070,933 (currency unspecified)
- Average interest rate: 12.05%
- Average DTI: 13.33%

Interpretation note: total payments received can include principal, interest and other amounts; it should not be treated as profit. These values are reported from the workbook summary, not independently audited against source systems.

5. Data dictionary and terminology

Field	Meaning / caution

id	Loan record identifier; check uniqueness before counting applications.
member_id	Borrower/member identifier; potentially sensitive and not needed in public sample data.
address_state	US state abbreviation associated with the borrower.
application_type	Application arrangement, such as individual or joint where present.
emp_length	Reported length of employment.
emp_title	Reported employer or job title; may contain identifying information.
grade	Lending risk grade assigned in the source data.
sub_grade	More detailed lending risk grade.
home_ownership	Reported housing status, such as rent, own or mortgage.
issue_date	Date the loan was issued.
last_credit_pull_date	Date of the most recent recorded credit check.
last_payment_date	Date of the most recent recorded payment.
next_payment_date	Scheduled next payment date in the source extract.
loan_status	Recorded loan outcome or current state, such as Fully Paid, Current or Charged Off.
Good Loan vs Bad Loan	Derived classification in the workbook; inspect the formula before treating it as a formal credit-risk definition.
purpose	Borrower-stated loan purpose.
term	Contractual loan duration, e.g., 36 or 60 months.
verification_status	Income or information verification status.
annual_income	Reported annual borrower income; currency unspecified.
dti	Debt-to-income ratio, generally debt obligations relative to income; confirm source-specific methodology.
installment	Scheduled installment amount.
int_rate	Loan interest rate as a decimal value, formatted as a percentage.
loan_amount	Original loan principal amount.
total_acc	Number of credit accounts recorded for the borrower.
total_payment	Cumulative payments received as represented in the extract.

6. KPI definitions

- Total loan applications: Count of loan records (or distinct id values after confirming id uniqueness).
- Total funded amount: Sum of loan_amount.
- Total amount received: Sum of total_payment.
- Average interest rate: Arithmetic mean of int_rate; this is not a balance-weighted yield.
- Average DTI: Arithmetic mean of dti; verify how zero, missing or extreme values are handled.
- Good loan percentage: Number classified Good Loan / total classified loans × 100.
- Bad loan percentage: Number classified Bad Loan / total classified loans × 100.
- MTD / PMTD: Month-to-date / previous-month-to-date, using a clearly defined reporting cutoff date.

7. Suggested analysis questions

- How do monthly loan application counts and funded amounts change?
- Which loan purposes, states and terms account for the largest funded volumes?
- How do interest rates and DTI vary across borrower and loan segments?
- What share of loans are current, fully paid or charged off?
- Does the derived good/bad classification change across loan grades and employment-length groups?

8. Data quality, assumptions and limitations

- Check duplicate id and member_id values and missing fields before interpreting counts.
- Validate date types, negative or outlier amounts, and rates stored as decimals.
- Review the good/bad formula and document treatment of Current loans.
- Loan status is a snapshot; avoid treating the data as a time-to-default model.
- Loan issue dates and payment dates are different timelines; payments received are cumulative.
- Do not infer causal drivers or make lending decisions from dashboard correlations.
- Confirm source license and anonymization; remove borrower identifiers and employer details before publishing a public dataset.

9. Tools and deliverables

Microsoft Excel: structured loan dataset, formulas, pivot-style KPI summaries and dashboard visualizations. GitHub deliverables: the workbook (only if publication rights and privacy checks permit), this project documentation, README.md, and optional screenshots of the two dashboards.

10. Portfolio description

Developed an Excel-based bank loan dashboard using 38,576 loan records to report funding, payment activity, interest rates, debt-to-income ratios and loan outcomes. Organized key lending measures into summary and overview dashboards to support portfolio monitoring and segmented analysis.